Exponential time hypothesis is false... but the verified algorithm is slower than trying all 2^n assignments at every practical size.

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November 5, 2026

Abstract

Placeholder. Item 09 writes the abstract (under 200 words; no number without a section that regenerates it). The title's sentence made precise: what is verified is a guarantee with existential constants and the exponent $n(\log\log n)^2/\sqrt{\log n}$, whose factor exceeds 1 for every $50 \le n < e^{5504}$, so the verified guarantee is weaker than the trivial $2^n \cdot \text{poly}$ guarantee at every practical size; the machine's running time is unknown because the constants are never computed; the manuscript's sharper exponents are unverified; the statement does refute ETH; the audit found no flaw.

1 Introduction

Placeholder. Item 09 drafts this section: the title's sentence made precise in the first paragraph (a guarantee, not a running time; existential constants never computed; $\rho(n) > n$ for every $50 \le n < e^{5504}$; the sharper exponents unverified); the one-paragraph summary; what is claimed and what is not; the stance (a careful reading by people who checked; no flaw found). Ends with what this does not claim".

2 The verified statement, read exactly

Placeholder. Item 02 drafts this section: the comparator's definitions one by one with their lines, the quantifier order, the encoding length, the exponent `runtimeScale`; the table (label `tab:runtime-scale`) and figure (label `fig:runtime-scale`) of $\rho(n)/n$ (from `paper3/scripts/runtime\_scale.py`); the title's claim stated precisely. Ends with what this does not claim".

3 What the manuscript claims beyond the verified statement

Placeholder. Item 03 drafts this section: Theorem 1.1's two sharper exponents, Corollary 1.4's quantitative circuit bound, Appendix C's consequences, Appendix D, each tagged verified or paper-only with page and line of the extract. Ends with what this does not claim".

4 The proof, inspected

Placeholder. Item 04 drafts this section: the closure size, the zero counts, the 33 identical definitions, the compile lemmas concluding in the comparator's own `Trace`, the unpatched Mathlib, the absence of a recorded axiom audit, of CI and of a second kernel; joined with the replay of Section 5. Ends with what this does not claim".

5 Independent replay

Placeholder. Item 01 drafts this section: the sandboxed build of the SubexponentialSAT closure from a fresh copy of the clone, the `#print axioms` output, the Lean and Mathlib versions, the time, the hashes of the logs, and whether nanoda ran. Ends with what this does not claim".

6 What follows

Placeholder. Item 05 drafts this section: which hypotheses the verified statement refutes (ETH, SETH, Gap-ETH, #ETH, NSETH, the randomised forms) and how; in general terms, that a lower bound derived from ETH or SETH loses its premise while a statement "no $(c-\varepsilon)^k$ algorithm" is not itself refuted (the manuscript's §1.1.3); no example from any width parameter. Ends with what this does not claim".

7 For the practitioner

Placeholder. Item 06 drafts this section: a SAT solver and a model counter on random 3-CNF at $n = 20 \ldots 60$ against $2^{\rho(n)}$ and 2^n (figure labelled `fig:practitioner`, from `paper3/scripts/practitioner.py`); the verified guarantee is never the binding one on any machine. Ends with what this does not claim".

8 The statement in our Lean

Placeholder. Item 07 drafts this section: the restatement `SubexpCount` over the verbatim machine model under `lean/MOSPFormalization/ETH/`, the axiom audit, the byte-level rediff against the comparator file; no theorem of ours about the hypothesis is proved, and none is claimed. Ends with what this does not claim".

9 Related work and context

Placeholder. Item 08 drafts this section: the human audit of the August 2026 results (arXiv 2608.14673), the working group named in the collection's history, Impagliazzo–Paturi, Impagliazzo–Paturi–Zane, the comparator mechanism, nanoda; "we found no prior statement", never "first". Ends with what this does not claim".

10 Conclusion

Placeholder. Item 09 drafts this section: what is claimed and what is not; every entry of `paper3/ETH\_AMBIGUOUS.MD` that affects the text, listed as open by number. Ends with what this does not claim".

Data, code and proofs

Placeholder. Item 11 completes this statement: the repository, the scripts under `paper3/scripts/` that regenerate every number, the replay's commands and log hashes under `paper3/replay/`, the Lean names and the axiom audit under `paper3/lean/`, and the read-only clone of https://github.com/openai/math at commit `4eaee875`.

Acknowledgements

Placeholder. Item 11 writes the acknowledgements, with the disclosure of AI use in the form of `paper1/latex/sec8\_ack.tex`.

References

Appendix A Formal names

Every Lean name the paper cites, with its file under `lean/MOSPFormalization/ETH/`; the axiom audit is `paper3/lean/axiom_check.lean` with its committed log.

	result	Lean names
A.1	The comparator statement restated as a `Prop`, `SubexpCount` k, over the verbatim copy of the machine model; never proved, never an axiom	`MOSPFormalization.SubexpPathwidth.SubexpCount`
A.2	$\rho(n) = o(n)$: the verified exponent is subexponential in the limit	`MOSPFormalization.SubexpPathwidth.runtimeScale\_isLittleO`

Placeholder. Item 07 moves these declarations to `lean/MOSPFormalization/ETH/` and rewrites the names above to their new namespace.